Park overview: energy, cost and machine states

Simulated park of 10 laser-sintering 3D printers in two halls. Energy from the meters' kWh counters; cost with an assumed time-of-use tariff (peak 0.24 €/kWh on weekdays 07–20 h, otherwise 0.17 €/kWh).

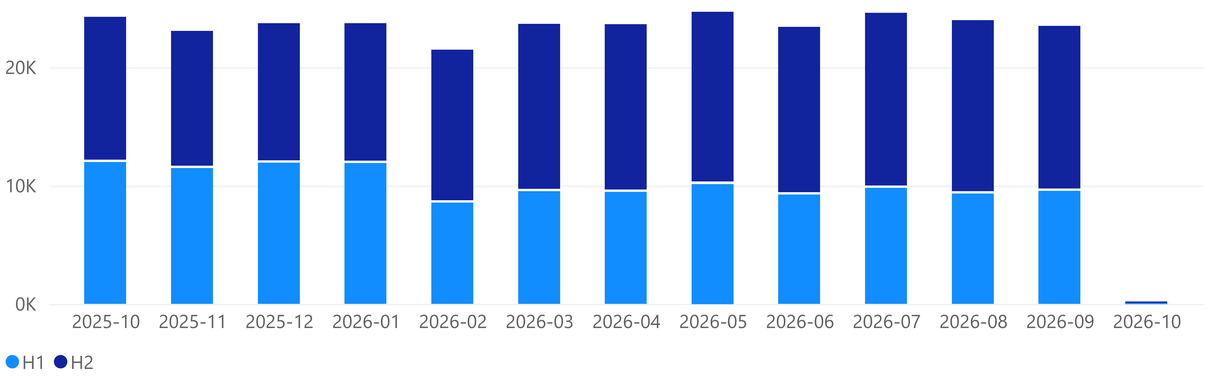
Hall: All

Date: 9/30/2025 – 10/8/2026

Energy (kWh)	Energy cost	Average price per kWh	Energy at peak tariff	Idle share of time	Data completeness
283,950	55,988 €	0.197 €	38.8%	17.4%	97.7%

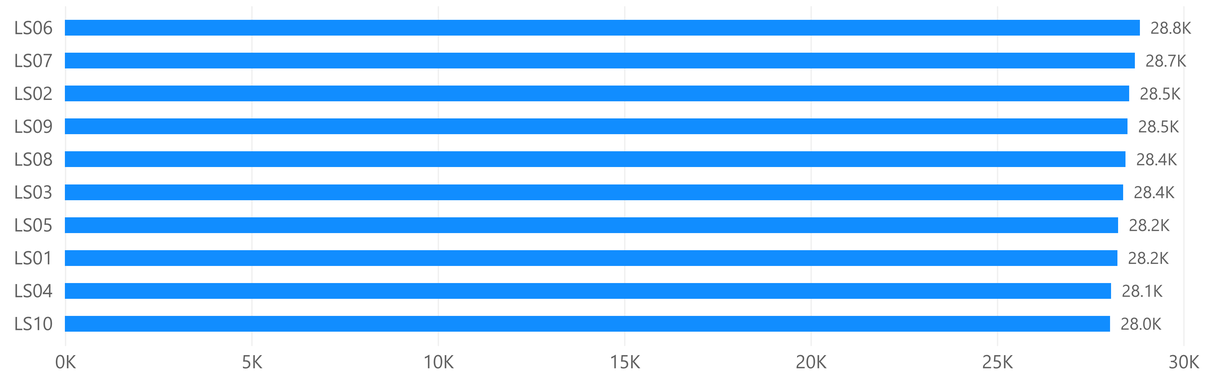
Energy per month, by hall (kWh)

Energy (kWh) by Month and Hall



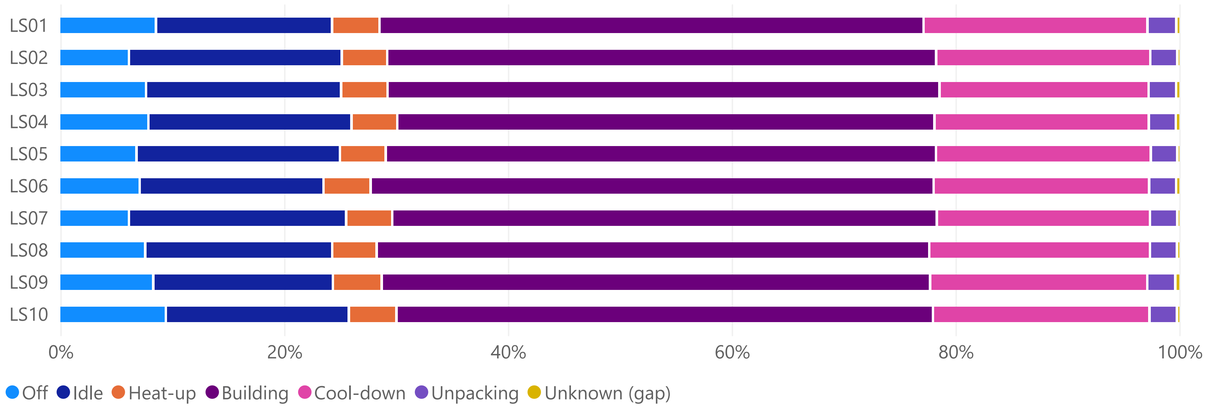
Energy by 3D printer (kWh)

Energy (kWh) by 3D printer



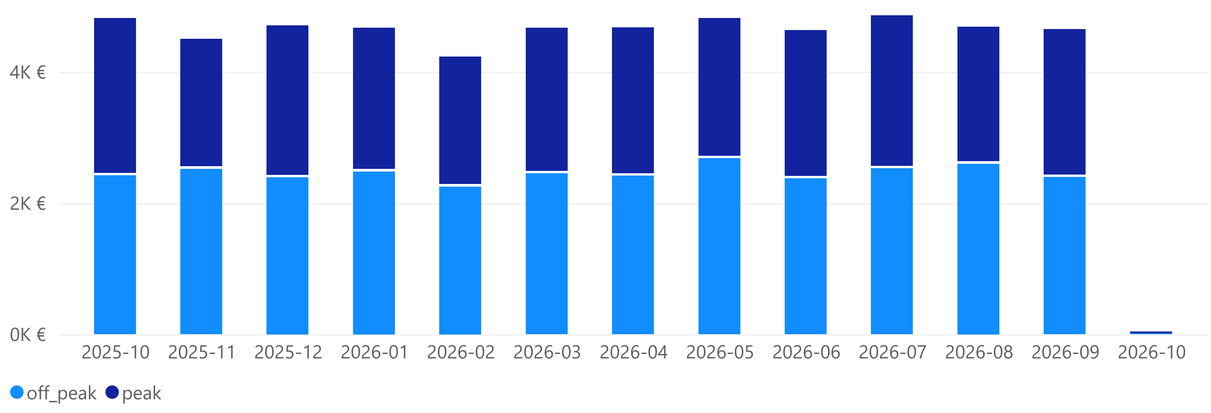
Share of time per machine state (unknown = data gap)

Hours by 3D printer and State



Energy cost per month, by tariff band

Energy cost by Month and Tariff



Machine drill-down: one 3D printer at a time

Pick a 3D printer at the top right (LS05 moved from hall 1 to hall 2 on 2026-02-01). Jobs and heat-ups that span a data gap are excluded.

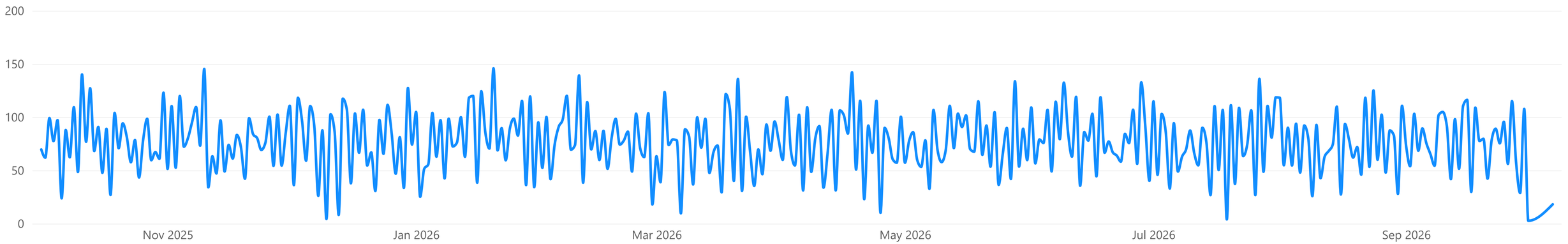
3D printer: LS05

Date: 9/30/2025 – 10/8/2026

Energy (kWh)	Build jobs	kWh per job	kWh per build hour	Idle share of time	Heat-up overruns
28,249	165	148.7	4.73	18.2%	7

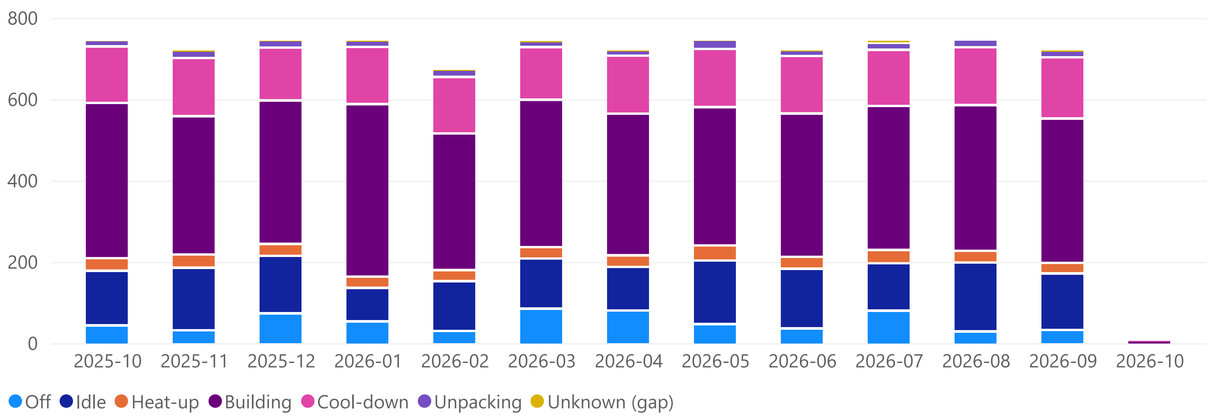
Energy per day (kWh)

Energy (kWh) by Day



Hours per machine state, by month

Hours by Month and State



Build jobs, most recent first

Job	Build start	Build hours	Job energy (kWh)	kWh per build hour
LS05-20260929T1857	2026-09-29	22.9	137.2	5.14
LS05-20260926T1947	2026-09-26	35.1	195.9	4.79
LS05-20260925T0204	2026-09-25	18.1	106.1	4.37
LS05-20260923T1121	2026-09-23	22.2	151.6	5.37
LS05-20260921T2320	2026-09-22	11.2	86.3	4.45
LS05-20260920T1312	2026-09-20	13.2	89.2	4.89
LS05-20260918T0020	2026-09-18	36.1	185.6	4.44
LS05-20260915T0507	2026-09-15	37.6	225.6	5.39
LS05-20260913T0047	2026-09-13	28.3	139.8	3.98
LS05-20260911T0721	2026-09-11	13.2	89.8	4.65
Total		**4,131.8**	**24,538.1**	**4.73**

Build jobs: energy per job and per build hour

Simulated park of 10 laser-sintering 3D printers. Job energy includes the heat-up before and the cool-down after the build. Jobs that span a data gap are excluded.

Hall

All

Date

9/30/2025 10/8/2026

Build jobs
1,622

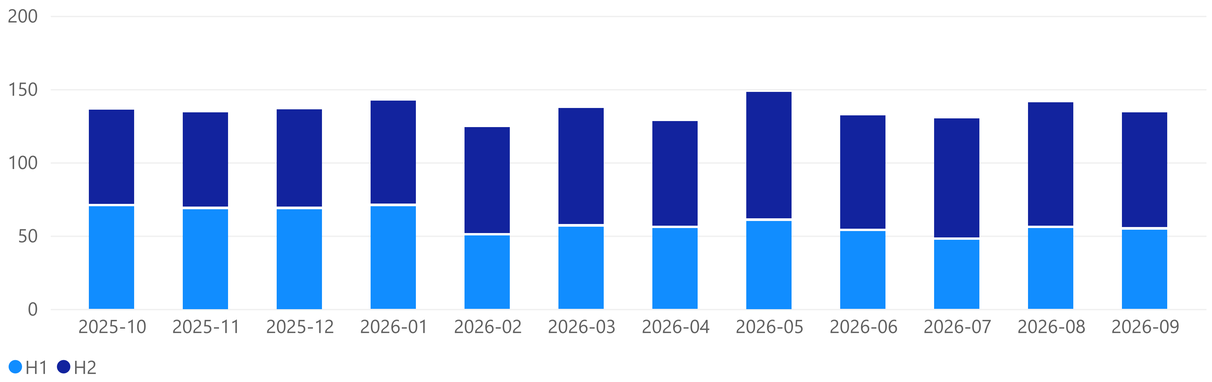
Build hours
40,427.7

kWh per job
150.2

kWh per build hour
4.79

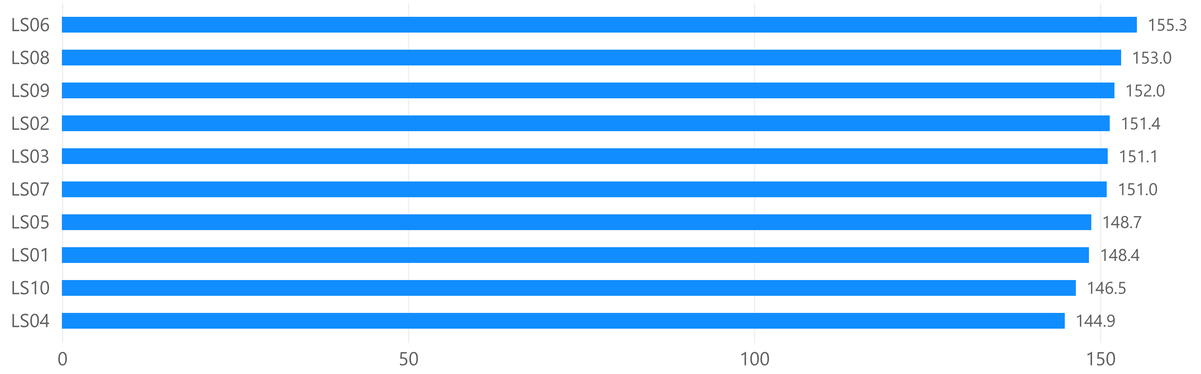
Build jobs per month, by hall

Build jobs by Month and Hall



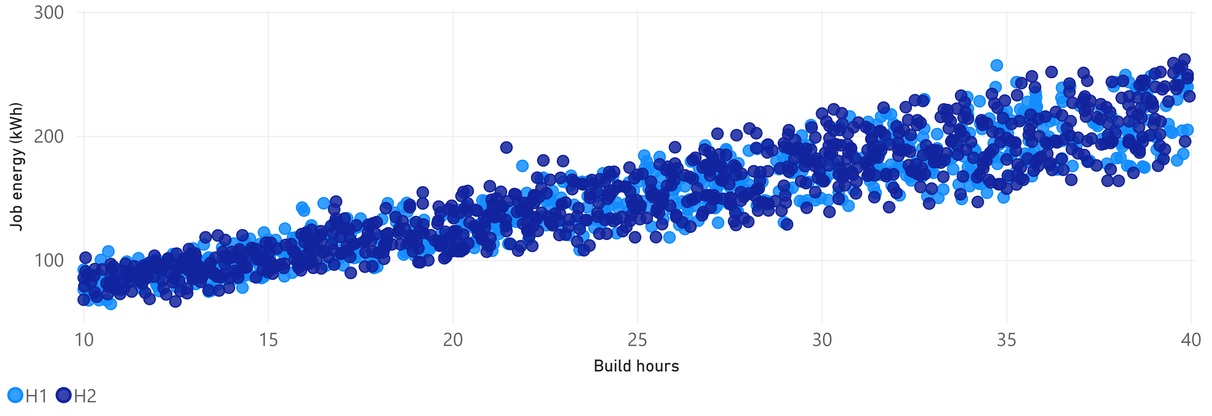
Average energy per job by 3D printer (kWh)

kWh per job by 3D printer



Each build job: build hours vs. job energy

Build hours and Job energy (kWh) by Job and Hall



Build jobs, largest energy first

Job	3D printer	Build start	Build hours	Job energy (kWh)	kWh per build hour
LS07-20260421T2209	LS07	2026-04-22	39.8	261.7	5.59
LS07-20260606T1718	LS07	2026-06-06	39.5	258.7	5.37
LS03-20260923T1826	LS03	2026-09-23	34.8	257.0	5.47
LS06-20260528T0654	LS06	2026-05-28	39.8	256.3	4.90
LS06-20260706T0317	LS06	2026-07-06	39.7	254.5	5.73
LS02-20260928T0739	LS02	2026-09-28	39.7	253.1	5.30
LS04-20260313T0009	LS04	2026-03-13	39.7	252.7	5.49
LS05-20260417T1303	LS05	2026-04-17	36.2	251.6	5.92
LS08-20260702T2343	LS08	2026-07-03	39.2	251.6	5.67
LS09-20251115T1751	LS09	2025-11-15	39.5	250.8	5.80
Total			**40,427.7**	**243,620.0**	**4.79**

Heat-up anomalies: the 2019 alert

Overrun = heat-up longer than 2.6 h (1.3 × the 2.0 h baseline). Heat-ups that span a data gap can't be judged and are excluded.

Hall

All

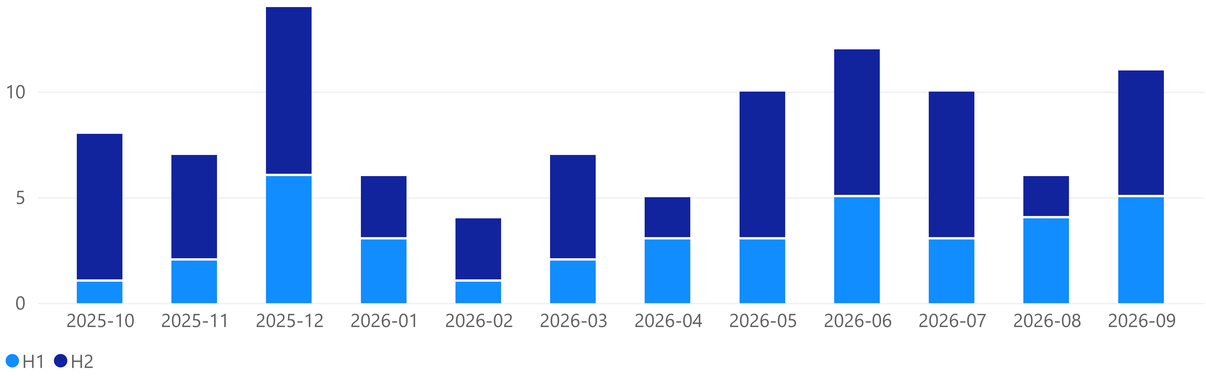
Date

9/30/2025 10/8/2026

Heat-ups	Overruns	Overrun rate	Average heat-up (h)
1,713	100	5.8%	2.12

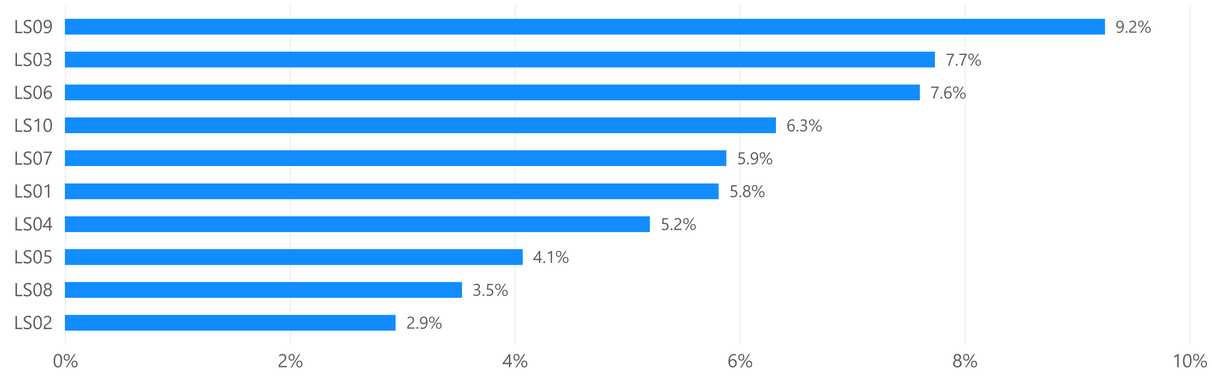
Heat-up overruns per month, by hall

Overruns by Month and Hall



Overrun rate by 3D printer

Overrun rate by 3D printer



Overrun heat-ups, most recent first

3D printer	Start (UTC)	Following build job	Duration (h)	Threshold (h)	Heat-up energy (kWh)
LS09	2026-09-30 04:23	LS09-20261006T1243	3.55	2.60	28.0
LS02	2026-09-28 04:29	LS02-20260928T0739	3.18	2.60	25.4
LS09	2026-09-24 01:47	LS09-20260924T0438	2.87	2.60	22.6
LS03	2026-09-23 12:48	LS03-20260923T1826	5.65	2.60	42.5
LS04	2026-09-22 20:38	LS04-20260923T0131	4.90	2.60	31.6
LS03	2026-09-16 12:59	LS03-20260916T1703	4.08	2.60	30.4
LS07	2026-09-12 17:29	LS07-20260912T2203	4.58	2.60	39.9
LS08	2026-09-11 14:24	LS08-20260911T1907	4.73	2.60	34.1
LS01	2026-09-06 11:29	LS01-20260906T1630	5.03	2.60	35.6
LS06	2026-09-05 08:39	LS06-20260905T1154	3.27	2.60	22.5
Total			**3.97**		**2,958.3**

Data quality: what arrived versus what was expected

Completeness = minutes with a valid reading ÷ expected minutes. October 2026 only has a few hours of live data.

Hall

All

Date

9/30/2025 10/8/2026

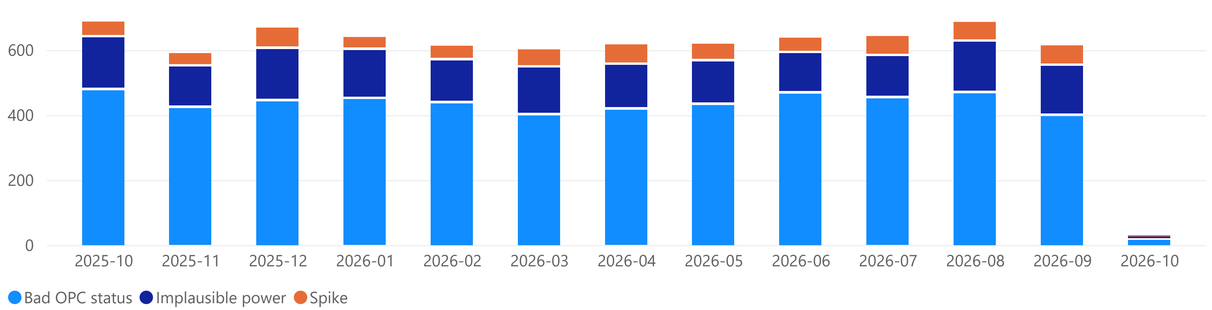
Completeness	Events received	Duplicates removed	Rejected readings	Late events	Counter discontinuities
97.7%	5,265,050	10,554	7,652	3,715	10

Completeness by 3D printer and month (red ≤ 80 %, yellow 95 %, green 100 %)

3D printer	2025-10	2025-11	2025-12	2026-01	2026-02	2026-03	2026-04	2026-05	2026-06	2026-07	2026-08	2026-09	2026-10	**Total**
LS01	99.5%	99.6%	99.5%	99.5%	99.2%	99.7%	99.6%	99.2%	99.4%	99.4%	99.4%	99.4%	3.4%	**97.6%**
LS02	99.7%	99.7%	99.7%	99.5%	99.8%	99.5%	99.5%	99.3%	99.6%	99.2%	99.2%	99.9%	3.4%	**97.7%**
LS03	99.7%	99.0%	99.7%	99.3%	99.5%	99.6%	99.8%	99.4%	99.4%	99.0%	99.6%	98.9%	3.4%	**97.6%**
LS04	99.4%	99.5%	99.5%	99.6%	99.4%	98.5%	99.2%	99.4%	99.6%	99.3%	99.8%	99.6%	3.4%	**97.6%**
LS05	99.7%	99.4%	99.7%	99.5%	99.6%	99.4%	99.5%	99.8%	99.6%	98.8%	99.9%	99.5%	3.4%	**97.7%**
LS06	99.3%	99.4%	99.4%	99.9%	99.9%	99.1%	99.1%	99.6%	99.1%	99.6%	99.1%	99.8%	3.4%	**97.6%**
LS07	99.4%	99.3%	99.5%	99.9%	99.2%	99.1%	99.5%	99.9%	99.7%	99.4%	99.7%	99.9%	3.4%	**97.7%**
LS08	99.1%	99.3%	99.8%	99.5%	99.5%	99.6%	99.1%	99.5%	99.9%	99.7%	99.7%	99.2%	3.4%	**97.7%**
LS09	99.7%	99.9%	99.4%	99.2%	99.1%	99.7%	98.6%	99.1%	99.4%	99.5%	99.1%	99.7%	3.4%	**97.6%**
LS10	99.6%	99.9%	99.5%	99.5%	98.9%	99.5%	99.5%	99.2%	99.5%	99.3%	99.9%	99.9%	3.4%	**97.7%**
Total	**99.5%**	**99.5%**	**99.6%**	**99.5%**	**99.4%**	**99.4%**	**99.3%**	**99.4%**	**99.5%**	**99.3%**	**99.5%**	**99.6%**	**3.4%**	**97.7%**

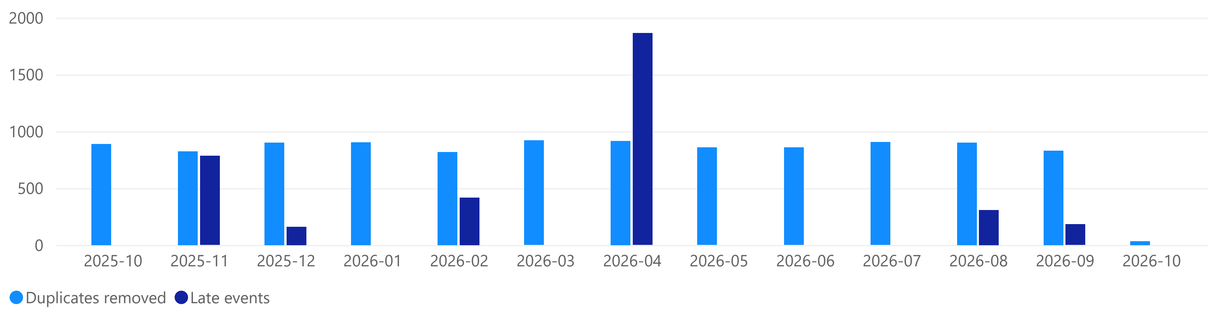
Rejected readings per month, by reason

Bad OPC status, Implausible power and Spike by Month



Duplicates removed and late events per month

Duplicates removed and Late events by Month



Market prices: does the park buy at expensive hours?

Hourly day-ahead prices for the Swiss bidding zone (Energy-Charts; data Bundesnetzagentur | SMARD.de, CC BY 4.0) against the simulated consumption. Energy component only: the fixed tariff also covers grid fees and levies.

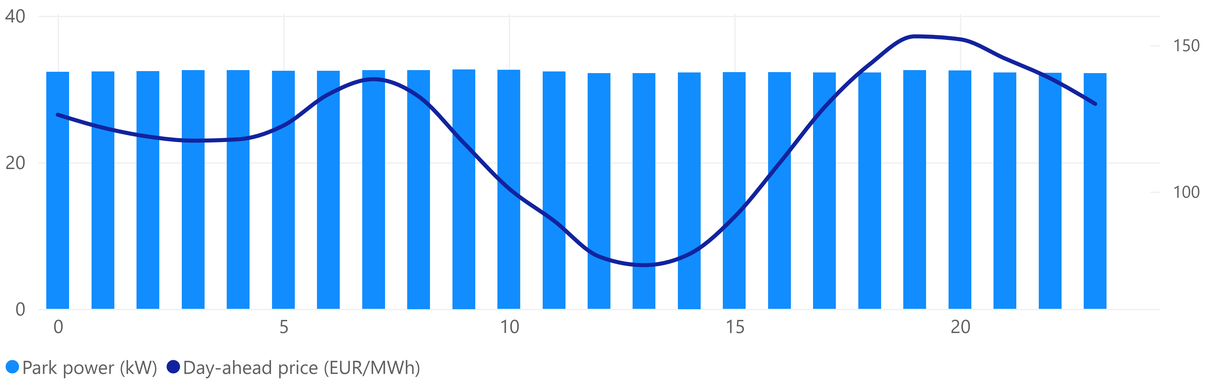
Hall: All

Date: 9/30/2025 – 10/8/2026

Park price (EUR/MWh)	Market average (EUR/MWh)	Load-profile premium	Energy at day-ahead prices	Negative-price hours	kWh at negative prices
119.4	119.2	+0.2%	33,901 €	239	7,942

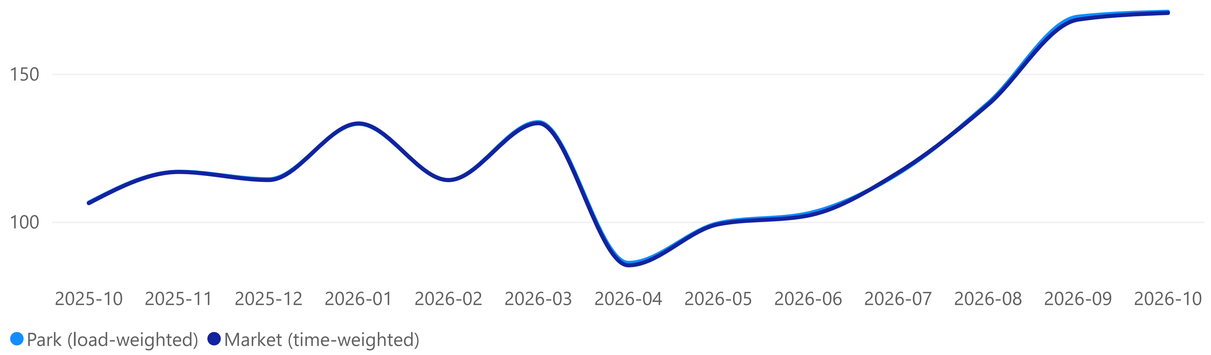
Park power and day-ahead price by hour of day

Park power (kW) and Day-ahead price (EUR/MWh) by Hour of day (local)



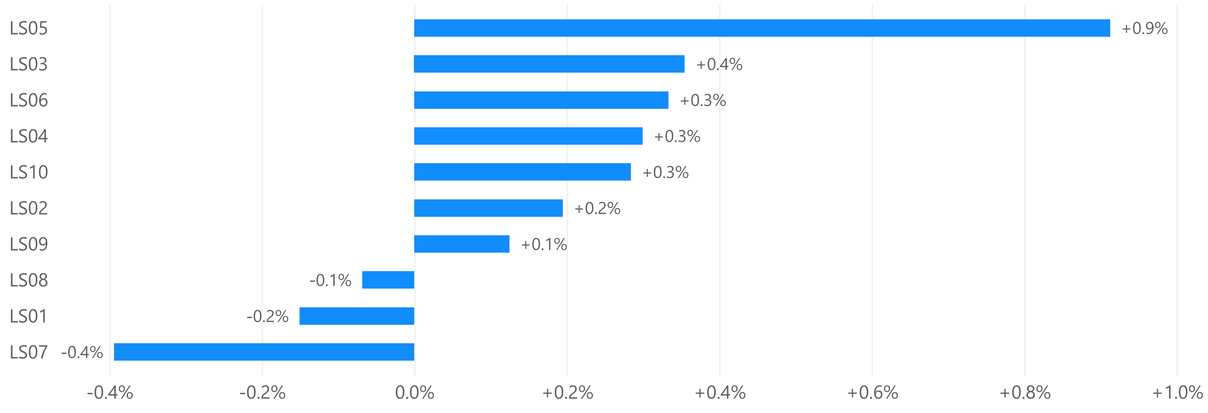
Park price and market price by month (EUR/MWh)

Park (load-weighted) and Market (time-weighted) by Month



Load-profile premium by 3D printer

Premium vs market by 3D printer



Hours with a negative day-ahead price by month

Negative-price hours by Month

